Semirings for database provenance

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What is a database?

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Loans

Loan_id	Member_id	Book_id	Borrow_date	Duration
43666	4221	389	17-02-2026	7
43666	4221	450	17-02-2026	7
43952	4222	975	17-02-2026	30
54673	4223	389	06-03-2026	14
58930	4224	975	09-03-2026	30
58930	4224	879	09-03-2026	30

Books

Book_id	Book_title	Genre	Copies
389	Nineteen Eighty-Four	Fiction	5
450	Brave New World	Fiction	4
879	Divine Comedy	Poetry	6
975	Critique of Pure Reason	Philosophy	3

Members

Member_id	First_name	Last_name
4221	Aditya	Prakash
4222	Graham	Leigh
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4224	Luigi	Brunetta

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$\{(\textbf{Member\_id}, 4221), (\textbf{First\_name}, \text{Aditya}), (\textbf{Last\_name}, \text{Prakash})\} \in$ *Members*

How do we ask questions?

With queries!

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In relational algebra, a query takes a database, applies transformations, and outputs a relation.

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The *relational algebra* consists of the following operations:

- Selection, $\sigma_\theta(R)$, filters tuples.
- Projection, $\pi_L(R)$, removes attributes.
- Renaming, $\rho_\beta(R)$, renames attributes.
- Natural join, $R \bowtie S$, merges relations on common attributes.
- Union, $R \cup S$, set-theoretic union.
- Difference, $R - S$, set-theoretic complement.

Example query

Query: Find the title of every book borrowed for 7 or 14 days.

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$$\pi_{Book\_title}\left(\sigma_{Duration=7}(Loans \bowtie Books) \cup \sigma_{Duration=14}(Loans \bowtie Books)\right)$$

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We represent it by tagging tuples with annotations.

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58930	4224	879	09-03-2026	30	l_6

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We define:

- An *S-relation* as a function $R : \mathbb{D}^U \to S$ with finite support.
- An *S-database* as a finite set of S-relations.

What happens to the annotations?

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Union adds together the annotations, while join multiplies them.
We also use 0 to indicate absence, while 1 indicates the presence of a tuple.
Together, these give us the algebraic structure $(S, +, \cdot, 0, 1)$.

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Book_title
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Brave New World

What happens to the annotations?

$$\pi_{Book\_title}\Big(\sigma_{Duration=7}(Loans \bowtie Books) \cup \sigma_{Duration=14}(Loans \bowtie Books)\Big)$$

Book_title	
Nineteen Eighty-Four	$l_1 \cdot b_1 + l_4 \cdot b_1$
Brave New World	$l_2 \cdot b_2$

When do semirings come in?

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For all relations R, S, T over attribute set U:

$$
\begin{aligned}
R \cup (S \cup T) &= (R \cup S) \cup T \\
R \cup S &= S \cup R \\
R \cup \mathbf{0}_U &= R \\
R \bowtie (S \bowtie T) &= (R \bowtie S) \bowtie T \\
R \bowtie S &= S \bowtie R \\
R \bowtie \mathbf{1} &= R \\
R \bowtie (S \cup T) &= (R \bowtie S) \cup (R \bowtie T) \\
R \bowtie \mathbf{0}_U &= \mathbf{0}_U
\end{aligned}
$$

What is a (commutative) semiring?

An algebraic structure $(S, +, \cdot, 0, 1)$ *such that for all* $a, b, c \in S$*:*

$$a + (b + c) = (a + b) + c$$
$$a + b = b + a$$
$$a + 0 = a$$
$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$
$$a \cdot b = b \cdot a$$
$$a \cdot 1 = a$$
$$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$$
$$a \cdot 0 = 0$$

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a \cdot 0 &= 0
\end{aligned}
$$

Examples:

- $\mathbb{N} = (\{0, 1, 2, \dots\}, +, \times, 0, 1)$
- $\mathbb{B} = (\{0, 1\}, \vee, \wedge, \perp, \top)$

Extending the framework

Further research has focused on extending the framework beyond the positive relational algebra to more expressive queries.

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We briefly consider one case, aggregation.

Aggregate queries

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title	genre	price
Book 1	biography	40
Book 2	fantasy	50
Book 3	poetry	60
Book 4	biography	30
Book 5	fantasy	10
Book 6	poetry	20

Aggregate queries

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$\gamma_{genre, AVG(price)}(Books)$

genre	AVG(price)
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Semirings alone are insufficient

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Book 4	biography	30	b_4
Book 5	fantasy	10	b_5
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Book 4	biography	30	b_4
Book 5	fantasy	10	b_5
Book 6	poetry	20	b_6

Semimodules to the rescue

We need an operation where the values are scaled by the annotations.

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We need an operation where the values are scaled by the annotations.
That is exactly what scalar multiplication does!

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An *S-semimodule* $(M, +_M, 0_M, *)$ is a commutative monoid $(M, +_M, 0_M)$ with scalar multiplication $* : S \times M \to M$.

Selected contributions

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- Proved the relationship between relational algebra identities and the commutative semiring axioms.
- Proved the fundamental theorem and the universality of the polynomial semiring.
- Proved the insufficiency of standard semirings for modelling aggregation.